Power Budget: User Interface Board

Team Number:	103
Project Name:	**GripRx**
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Version:	**2 (Duotao Gao, 2026-10-29)**

A. List all major powered components, excluding power sources, regulators and passive elements.

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano (MCU board)	DM164150	1.8 - 5.5 V	1	350.0	350.0	mA
	Character LCD + backlight	NHD-0216K1Z-FSW-FBW-L	4.5 - 5.5 V	1	23.0	23.0	mA
	Quad op amp + speaker drive	MCP6004-I/P	1.8 - 6.0 V	1	30.0	30.0	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano (MCU board)	DM164150	1.8 - 5.5 V	1	350.0	350.0	mA
	Character LCD + backlight	NHD-0216K1Z-FSW-FBW-L	4.5 - 5.5 V	1	23.0	23.0	mA
	Quad op amp + speaker drive	MCP6004-I/P	1.8 - 6.0 V	1	30.0	30.0	mA
	Other small circuit loads	LEDs, buttons, pots, bias	5 V budget	1	13.3	13.3	mA
					Subtotal	416.3	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	520.3	mA
c2. Regulator or Source Choice	+5V linear regulator	L7805ABV-DG	7.5 - 18 V	1	1500.0	1500.0	mA
					Total Remaining Current Available on +5V Rail	979.7	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	9 V 3 A plug-in wall adapter, issued in class	BestCH 9 V 3 A (B09ZTKTLGW)	See adapter label	9 V	3000.0	3000.0	mA
Power Rails Connected to External Power Source 1	Regulator ground-current allowance	L7805ABV-DG	9 V input	1	6.0	6.0	mA
	+5V linear regulator	L7805ABV-DG	7.5 - 18 V	1	1500.0	1500.0	mA
			Total Remaining Current Available on External Power Source 1			1494.0	mA

Notes

All UI loads use +5 V. The board uses a wall-plug adapter. Battery life is not applicable.

Current basis: Nano 350 mA uses the MCU VDD-pin absolute rating (DS40002147D) as the budget limit, not normal board draw. MCP6004 uses the 30 mA output/supply-pin absolute rating (DS20001733L, p. 3). These are stress ratings, not normal operating currents. LCD 23 mA = 3 mA logic + 20 mA backlight maximum. Auxiliary loads total 13.25 mA. Speaker current is supplied by MCP6004 and is not added again.

Audio: U3A drives the speaker through C14 (220 uF) and R16 (330 ohm). A 2.5 V excursion / 330 ohm gives about 7.6 mA; a full 5 V excursion gives 15.2 mA. The 30 mA supply-pin rating is counted once for the chip, not four times for its four amplifiers. Confirm actual current and volume during testing.

Budget load = 416.25 mA. Including 25% margin = 520.31 mA. Estimated regulator dissipation is about 2.14 W at 9 V input, including a 6 mA ground-current allowance. Provide a suitable heatsink and verify regulator temperature; the 1.5 A rating requires adequate cooling.